

Professional Summary

PhD in Mathematical Models and Methods in Engineering, with strong expertise in C++20, CUDA, ROCm, and libraries for HPC on CPUs and GPUs. Extensive background in developing performance-portable scientific software across a wide range of software libraries and hardware vendors, including NVIDIA and AMD GPUs, as well as in optimisation and benchmarking within Linux-based distributed environments. Passionate about accelerators, heterogeneous and distributed computing, physics, research, teaching and dissemination. Implemented advanced GPU simulation codes in C++20/CUDA-C and foundational deep-learning models in C++14 with Eigen. Master's degrees in Applied Mathematics and Nuclear Engineering. Demonstrated ability in R&D software development, including code reviews and issue-driven methodologies within Git-based workflows. Strong communication and teamwork skills, further developed through experience in teaching, especially on numerical methods via advanced C++, industry collaborations, co-organising scientific mini-symposia, and delivering scientific talks. Recently began acting as a co-supervisor for Master's and PhD theses on HPC topics (Thrust vs PETSc-on-GPU, Alpaka vs CUDA-C for matrix operations, mixed precision on GPUs). Currently Technical Manager at the MOX Laboratory – Modelling and Scientific Computing – PoliMi; Linux HPC system configuration (HW, OS) and deployment (openPBS, spack, Apptainer).

Education

- 11/2021 – 02/2025 **PhD in Mathematical Models and Methods in Engineering**, *Laboratory for Modelling and Scientific Computing (MOX)*, Mathematics Department, Politecnico di Milano
 Design and development of C++20 code for GPUs and CPUs (Thrust, std::execution/TBB, CUDA, HIP).
PhD Thesis: "Material Point Method for Compressible Flows: a Portable, Massively Parallel Implementation for High-Performance Computing Architectures"
 Supervisor: prof. Carlo de Falco, carlo.defalco@polimi.it
- 2021 **National Habilitation**, Senior Nuclear Engineer
- 10/2020 – 10/2021 **MSc in Nuclear Eng.**, Nuclear System Physics, *Politecnico di Milano*, 107/110
- 09/2017 – 10/2020 **MSc in Math. Eng.**, Computational Sciences and Engineering, *Politecnico di Milano*, 110/110
- 09/2013 – 07/2017 **BSc in Ingegneria Matematica**, *Politecnico di Milano*, 110/110

Professional Experience

- 12/2023 – present (Permanent Position) **Technical Manager - MOX Laboratory for Modelling and Scientific Computing**, Mathematics Department, Politecnico di Milano
- Lead the design, implementation, and maintenance of distributed computing systems tailored for scientific research, ensuring optimal performance and reliability in high-demand environments.
 - Development and optimisation of scientific software, with a strong emphasis on modern C++, GPU porting, refactoring, testing and integration with AI frameworks, while fostering productive collaborations between academia and industry.
 - PI of the "MatBa" project, providing access to CINECA HPC resources for DMAT researchers, as well as finding optimal solutions specific to scientific computing needs.
 - Lead a project to renew an outdated HPC infrastructure, securing funding to acquire modern, energy efficient and highly parallel servers.
 - Supervision of Technical Staff for Linux System Administration.
 - Co-supervision of MSc/PhD students on GPU/HPC software (PhD: gasdynamics/FSI with Thrust & PIC-MPM on clusters; MSc project: CUDA-C vs Alpaka for matrix ops; MSc thesis: PETSc-on-GPU for finite-difference gasdynamics, GPU-aware mixed precision optimisations, [nanobind](#) data sketches).
- 2022 – present **Teaching Assistant - Scientific Computing MSc Courses**, Politecnico di Milano
- Delivered advanced C++/numerical labs at Master's level — Runge-Kutta example: github.com/pacs-course/pacs-Labs/tree/main/Labs/2025/10-rkf-2
 - Covered technical topics such as C++23, MPI, OpenMP, Octave, Bash, Make, Docker/Podman, Singularity/Apptainer, Spack, gdb, lcov, valgrind, CMake, and pybind11, as well as C++ libraries such as Eigen, Boost and abstraction layers as NVIDIA and ROCm Thrust for performance portability
 - Managing graduate student teams, providing guidance and support to enhance their understanding of complex scientific computing concepts.

Key Technical Skills

Programming	C++20/23, CUDA-C, HIP, ROCm, MPI, Eigen, Boost, (roc)Thrust, std::execution/TBB, C, Make, gdb, lcov, valgrind, Nsight Systems, shell, Octave/Matlab, Python (scripting), pybind11 and Numba
Software Tools	git/GitHub, Docker/Podman, Singularity/Apptainer, spack, CMake, Doxygen, SLURM/PBS
Numerical Methods	FEM, DG, PIC/MPM, Numerical Linear Algebra, Symplectic Integrators, Runge-Kutta Methods
OS & Management	Linux (Debian, Arch and RHEL based systems), MS Windows, Virtualisation tools; iDRAC, iLO
Scientific research software usage	○ SOLPS-ITER, coupled Finite Volumes and Monte Carlo code for the simulation of boundary plasma in magnetic confinement fusion devices, including EIRENE Boltzmann-equation Monte Carlo solver; ○ PyFR, solver for advection–diffusion using Flux Reconstruction, designed for performance portability across multiple backends (CPU/GPU) via a Python based domain-specific templating layer.

Professional Development

Red Hat Courses	Ansible and Kubernetes (2024-2026).
NVIDIA Courses	Scaling Workloads Across Multiple GPUs with CUDA C++, GPU Acceleration with the C++ Standard Library, Building Container Images for HPC, Accelerated Computing with CUDA Python, Accelerating CUDA C Applications with Concurrent Streams, and more (2024-2025).
CINECA Courses	Build Systems and Package Managers in HPC, Parallel Computing on Traditional Core-Based and Emerging GPU-Based Architectures, Numerical Methods for Parallel CFD (2021-2023).
Conferences and Workshops	Continuous learning through attendance at various industry workshops, EuroHPC hackathons, and scientific conferences on HPC, AI, Scientific Machine Learning, Quantum Computing.

Publications and Conferences

- 2025 **Workshop Poster (selected)**, *Portable Parallel Computing: C++ Strategies for Cross-Architecture Performance*, Baioni P.J., Benacchio T., Capone L., de Falco C., [E4: AI, HPC & QUANTUM 2025](#)
- 2024 **Journal Paper**, *Portable, massively parallel implementation of a material point method for compressible flows*, Baioni P.J., Benacchio T., Capone L., de Falco C., *Computational Particle Mechanics*, Springer. DOI: [10.1007/s40571-024-00864-2](https://doi.org/10.1007/s40571-024-00864-2), full text <https://rdcu.be/d20u2>
- 2024 **Review activity**, Reviewer for the *Journal of Computational Particle Mechanics*, Springer
- 2023 **Proceedings of Ecomas - Particles 2023**, *GPUs Based Material Point Method for Compressible Flows*, Baioni P.J., Benacchio T., Capone L., de Falco C. VIII International Conference on Particle-Based Methods, DOI: [10.23967/c.particles.2023.026](https://doi.org/10.23967/c.particles.2023.026)
- 2023 **Proceedings of SIMAI 2023**, *MS08 - Particle-Based Methods In Applied And Industrial Sciences*, Baioni P.J., Crescenzo N., Fois M., Moreno Martinez L. The XVI biannual congress of SIMAI, link: <https://web.unibas.it/simai2023/BookOfAbstracts.pdf>
- 2023 **Proceedings of SIMAI 2023**, *Material Point Method for Compressible CFD on GPUs*, Baioni P.J., Benacchio T., Capone L., de Falco C., Pellegrino E. The XVI biannual congress of SIMAI

FOSS Software

- 2020 Development of a C++14 code for a numerical methods and programming university course project, implementing a Deep Neural Network (DNN) from scratch, relying on the C++ Standard Library and Eigen; code available at <https://github.com/pjbaioni/neural-net>

Professional Memberships

- 2023 – present Member of INdAM, National Scientific Computing Group (GNCS)
- 2023 – present Member of Italian Society for Applied and Industrial Mathematics (SIMAI)
- 2022 – present Enrolled in the Order of Engineers of the Metropolitan City of Milan

Languages

- Italian Native speaker
- English C1 (TOEIC 2016); teaching MSc courses in English since 2022